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RESEARCH INTERESTS

Non-genetic neuromodulation with photothermal nanoparticles · Multifunctional fiber neural probes by thermal drawing · Surface chemistry for membrane anchoring and cell-type targeting · Memory engram allocation and control · Experience-independent concept transfer

EDUCATION

Sep 2020 — Aug 2026

Ph.D. in Brain and Cognitive Engineering

Korea Advanced Institute of Science and Technology (KAIST)

Integrated M.S.–Ph.D. program, advised by Prof. Seongjun Park · Thesis: Non-genetic Bimodal Neuromodulation using Fiber-based Neural Probe and Gold Nanorod

Mar 2015 — Aug 2020

B.S. in Electrical Engineering and Bio and Brain Engineering

Korea Advanced Institute of Science and Technology (KAIST)

Double major

RESEARCH EXPERIENCE

Sep 2026 — present

Postdoctoral Researcher

Medical Research Center, Seoul National University

Non-genetic engram allocation and control

- Allocating a hippocampal CA1 engram without genetic modification, using pulsed gold-nanorod photothermal stimulation delivered before contextual fear conditioning
- Suppressing retrieval of the allocated engram by continuous near-infrared irradiation during the memory test, with a no-pre-stimulation group establishing that the effect depends on the previously allocated population
- Running contextual fear conditioning with automated scoring of freezing and locomotor speed across allocation and non-allocation cohorts
- Maintaining transgenic mouse lines and recording population activity by fiber photometry in conditioned and neutral contexts
- Developing a relational framework in which engram identity is defined by overlap and co-firing with previously stored engrams, and testing experience-independent transfer computationally in a model of the entorhinal–dentate perforant pathway

Sep 2020 — Aug 2026

Doctoral Researcher
Bio and Neural Interfaces Laboratory, KAIST
Advised by Prof. Seongjun Park

- Built multifunctional fiber neural probes by thermal drawing, integrating a microfluidic channel, an optical waveguide and recording electrodes in one filament, and characterized their optical, fluidic, electrical and mechanical properties
- Synthesized gold nanorods and exchanged their surface ligands for cholesterol-terminated PEG to make them biocompatible and membrane-anchoring
- Carried out stereotaxic implantation and in vivo single- and multi-unit recording in mouse hippocampus and entorhinal cortex
- Established the photothermal stimulation parameters that separate inhibition from excitation, and assessed tissue response by NeuN and Iba1 immunohistochemistry
- Developed cell-type-specific non-genetic neuromodulation by conjugating antibodies to nanoparticles, validating targeting in primary neuronal culture by calcium imaging and immunostaining (project concluded when the laboratory relocated)

PUBLICATIONS

2026

Nongenetic in Vivo Bimodal Neuromodulation via Photothermal Gold Nanorods and a Multifunctional Fiber Neural Probe
C. Yoon, Y. Lee, G. Yim, S.-W. Lee, W. Lee, Y.-G. Park, S. Park
ACS Nano 20(28), 20001–20015
doi:10.1021/acsnano.6c02201

2026

Thermally Drawn Multifunctional All-Hydrogel Fibers for Anti-Fibrotic and Multimodal Neural Interfaces
C. Sung, K. S. Nam, Y. Kim, H. Kang, K. Kim, **C. Yoon**, S. Lee, A. Chung, J. Kang, Y.-G. Park, A. J. Park, H. Butt, H. Yuk, S. Park
Advanced Materials 38(9), e11634
doi:10.1002/adma.202511634

2022

A Personalized Electronic Tattoo for Healthcare Realized by On-the-Spot Assembly of an Intrinsically Conductive and Durable Liquid-Metal Composite
G.-H. Lee, H. Woo, **C. Yoon**, C. Yang, J.-Y. Bae, W. Kim, D. H. Lee, H. Kang, S. Han, S.-K. Kang, S. Park, H.-R. Kim, J.-W. Jeong, S. Park
Advanced Materials 34(32), 2204159
doi:10.1002/adma.202204159

2021

Recent Advances in Recording and Modulation Technologies for Next-Generation Neural Interfaces
J.-W. Hong, **C. Yoon**, K. Jo, J. H. Won, S. Park
iScience 24(12), 103550 · Review
doi:10.1016/j.isci.2021.103550

MANUSCRIPTS IN PREPARATION

In preparation

Non-genetic Engram Allocation and Manipulation via Gold Nanorod and Multifunctional Fiber-based Neural Probe
C. Yoon, et al.
Manuscript in preparation

In preparation

The GIVER: Geometric Induction of Vectorized Engram Readout for Experience-Independent Memory Transfer
C. Yoon, et al.
Computational study · Manuscript in preparation

CONFERENCE PRESENTATIONS

2026	**Non-Genetic Engram Allocation-and-Manipulation in Mouse Brain via Gold-Nanorod Mediated Photothermal Stimulation** Korean Society for Brain and Neural Sciences (KSBNS) Poster
2025	**Non-Genetic In Vivo Bimodal Neuromodulation Enabled by Fiber-Based Neural Probe and Gold Nanorod** Society for Neuroscience (SfN) Poster
2025	**Non-Genetic In Vivo Bimodal Neuromodulation Enabled by Fiber-Based Neural Probe and Gold Nanorod** IEEE International Conference on Neural Engineering (NER) Poster
2023	**Non-Genetic Bimodal In Vivo Neuromodulation Using Gold Nanorod-Mediated Photothermal Stimulation** Korean Society for Brain and Neural Sciences (KSBNS) Poster
2022	**Non-Genetic, Bidirectional In Vivo Photothermal Neuromodulation Enabled by Gold Nanorod and Fiber-Based Multifunctional Neural Interface** Korean BioChip Society (KBSC) Poster

SKILLS AND TECHNIQUES

Fabrication	Thermal drawing of multifunctional fibers, Preform design, Neural probe assembly
Nanomaterials	Seed-mediated gold nanorod synthesis, Ligand exchange (CTAB to cholesterol-PEG), Antibody conjugation for cell-type targeting, TEM, UV-Vis spectroscopy, zeta potential
In vivo	Stereotaxic implantation surgery, Single- and multi-unit electrophysiology, Fiber photometry, Photothermal neuromodulation, Intracranial viral delivery by nanojet injector and by Hamilton syringe with micropump (AAV-DJ CaMKII-GCaMP), Transcardial perfusion, brain sectioning, Immunohistochemistry (NeuN, Iba1, c-Fos)
Behaviour	Contextual fear conditioning, Automated scoring of freezing and locomotor speed with a custom DeepLabCut-based pipeline, Transgenic colony maintenance: mating and weaning of Thy1-GCaMP, Thy1-ChR2 and DAT-Cre lines
In vitro	Primary neuronal culture, Viral transfection, Calcium imaging, Culture perfusion and immunostaining
Characterization	Optical attenuation, Impedance spectroscopy, Bending stiffness, Infrared thermography
Computation	MATLAB, Python, SolidWorks, Bioheat modeling, Spike sorting and PETH analysis, DeepLabCut model training for pose estimation, Neural population and ensemble-overlap analysis